American Computer Science League

2022-2023 ● Contest 1: Short Problems ● Elementary Division

1. Computer Number Systems

Convert $1B8_{16}$ to base 10.

A. 335
B. 409
C. 430
D. 440
E. 450

2. Computer Number Systems

Which of the following has the most number of 1s in its binary representation?

ABE_{16} CDF_{16} 7531_8 6420_8

A. ABE_{16}
B. CDF_{16}
C. 7531_8
D. 6420_8
E. more than 1 answer

3. Computer Number Systems

The year 2022 is 3746 in base 8. How many octal years is it until the year 5072_8?

A. 1023
B. 1124
C. 1326
D. 2336
E. 10020

4. Computer Number Systems

Place the following numbers in increasing order:

a) 1101010_2 b) $5F_{16}$ c) 124_8 d) 100_{10}

A. b, c, a, d
B. b, c, d, a
C. c, b, d, a
D. c, d, a, b
E. d, a, c, b

5. Computer Number Systems

Compare each RGB component of the hexadecimal colors "#409AF7" and "#348DE9". What is the difference in hexadecimal for the component that has the smallest difference in value?

A. 0A
B. 0B
C. 0C
D. 0D
E. 0E

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1. 计算机数字系统

请将 $1B8_{16}$ 转换**为十进制数。**

A. 335
B. 409
C. 430
D. 440
E. 450

2. 计算机数字系统

下面**哪个数字转换成二进制数后，含有数字 1 的数量最多？**

ABE_{16} CDF_{16} 7531_8 6420_8

A. ABE_{16}
B. CDF_{16}
C. 7531_8
D. 6420_8
E. **不止一个**

3. 计算机数字系统

年份 2022 转换成八进制是 3746。一直到年份 5072_8 会有多少个可以转换成八进制的年份?

A. 1023
B. 1124
C. 1326
D. 2336
E. 10020

4. 计算机数字系统

请将以下数字从小到大进行排列：

a) 1101010_2 b) $5F_{16}$ c) 124_8 d) 100_{10}

A. b, c, a, d
B. b, c, d, a
C. c, b, d, a
D. c, d, a, b
E. d, a, c, b

5. 计算机数字系统

比较十六进制**数表示的**颜色 "#409AF7" 和 "#348DE9" 中每个 RGB **部分。差值**最小的**部分转换为**十六进制**后的**差是多少？

A. 0A
B. 0B
C. 0C
D. 0D
E. 0E